Google Data Analytics Capstone Project

Uber Ride Booking Data Analysis

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Tools Used

Python, SQL, Power BI

Introduction

Transportation services have become an essential part of daily life, and ride-sharing platforms such as Uber play a major role in meeting this demand. Every day, thousands of customers use the platform to book rides, while drivers rely on the system to connect with passengers efficiently.

The success of a ride-sharing platform depends on several factors, including ride completion rates, customer satisfaction, driver availability, and revenue generation. Understanding customer behavior and operational challenges can help businesses improve their services and make better decisions.

In this project, Uber ride booking data was analyzed using Python, SQL, and Power BI. The objective was to identify important trends in bookings, cancellations, customer retention, and revenue performance. The insights obtained from this analysis can help improve operational efficiency and enhance the overall customer experience.

Step 1: Ask

Business Background

Uber handles a large number of ride requests every day. However, not every booking results in a successful ride. Some rides are cancelled by customers, some by drivers, and others fail due to driver unavailability. These issues can reduce customer satisfaction and impact business revenue.

To better understand these challenges, an analysis was conducted on Uber booking data to identify patterns and opportunities for improvement.

Business Task

The main objective of this project is to examine Uber ride booking data and answer important business questions related to ride completion, cancellations, customer behavior, and revenue generation.

Business Questions

1. What percentage of rides are successfully completed?
2. How frequently are rides cancelled?
3. What factors contribute to booking failures?
4. Which customer groups generate the highest revenue?
5. How effectively does Uber retain customers?
6. What operational improvements can be recommended based on the findings?

Stakeholders

- Uber Operations Team
- Business Management Team
- Customer Experience Team
- Driver Management Team
- Data Analytics Team

Step 2: Prepare

Dataset Description

The dataset used in this project contains approximately 150,000 Uber booking records. It includes information related to ride requests, booking status, cancellation details, vehicle categories, customer activity, and revenue generated from rides.

Data Source

The dataset was collected from a publicly available source and used strictly for educational and analytical purposes.

Data Structure

The dataset includes information such as:

- Booking ID
- Booking Status
- Customer ID
- Vehicle Type
- Booking Value
- Pickup and Drop Locations
- Cancellation Details
- Ride Completion Information

Data Limitations

Although the dataset contains a large number of records, there are certain limitations:

- It represents only a sample of Uber operations.
- Geographic information is limited.
- External factors such as weather conditions and traffic data are unavailable.
- Results may not represent all regions where Uber operates.

Data Credibility Assessment

The dataset is suitable for learning and analytical purposes because it contains a large number of records and multiple business-related variables. However, since it is not directly sourced from Uber's internal systems, conclusions should be interpreted within the scope of the dataset.

Step 3: Process

Data Cleaning and Preparation

Before beginning the analysis, the dataset was carefully examined to ensure accuracy and consistency.

Several data preparation tasks were performed using Python.

Data Cleaning Activities

- Converted booking values into numeric format.
- Converted date fields into proper datetime format.

- Checked for missing values across all columns.
- Removed invalid records where necessary.
- Verified data types for each column.
- Filtered completed rides for revenue analysis.
- Standardized column formats for easier analysis.

Data Validation

To improve reliability, the dataset was reviewed for:

- Missing values
- Duplicate records
- Incorrect booking values
- Data type inconsistencies
- Invalid booking statuses

After cleaning and validation, the dataset was ready for further analysis using SQL and Python.

Step 4: Analyze

Overall Ride Performance

The first objective was to evaluate how efficiently rides were being completed.

Key Findings

- Around 60–62% of ride requests were successfully completed.
- Approximately 25% of bookings were cancelled.
- Nearly 13% of bookings failed due to operational reasons.

These figures indicate that a significant number of ride requests do not result in successful trips, highlighting opportunities for improvement.

Cancellation Analysis

Ride cancellations were analyzed to understand whether they were associated with specific time periods or operational conditions.

Findings

- Cancellation rates remained relatively stable throughout the day.
- Both drivers and customers contributed to cancellations.
- Cancellations were not concentrated within a single vehicle category.

This suggests that the cancellation issue is more operational than vehicle-specific.

Revenue Analysis

Revenue trends were examined to identify the most valuable segments of the business.

Findings

- Certain routes generated high revenue despite lower ride volumes.
- Revenue was not distributed evenly across all ride categories.
- Repeat customers contributed significantly more revenue than one-time users.

These findings indicate the importance of retaining existing customers and focusing on high-performing routes.

Customer Behavior Analysis

Customer activity was analyzed to better understand booking patterns and user engagement.

Customer Segments Identified

- One-Time Users
- Occasional Users
- Regular Users
- VIP Users

Findings

A large portion of customers used the service only once. However, customers who booked rides repeatedly generated the majority of total revenue.

This demonstrates that long-term customer retention is a key driver of business success.

Retention Analysis

Customer retention was examined to determine how frequently users returned after their first booking.

Findings

- Customer retention declined significantly after the first ride.
- Only a smaller percentage of users continued using the platform regularly.
- Retained customers generated substantially higher revenue.

These observations highlight the need for loyalty programs and customer engagement strategies.

Step 5: Share

Dashboard Development

To communicate findings effectively, an interactive Power BI dashboard was created.

The dashboard provides a visual representation of:

- Booking Status Distribution
- Ride Completion Rates
- Cancellation Trends
- Revenue Performance
- Customer Segmentation
- Driver Availability
- Route Analysis

Major Insights

Insight 1

Ride completion rates are lower than expected and represent a significant opportunity for operational improvement.

Insight 2

Customer and driver cancellations both contribute to service inefficiencies and revenue loss.

Insight 3

Most revenue is generated by returning customers rather than first-time users.

Insight 4

Driver shortages during peak demand periods increase booking failures.

Insight 5

A small number of routes contribute disproportionately to total revenue.

(Power BI dashboard screenshots can be inserted in this section.)

Step 6: Act

Based on the analysis, several recommendations can help improve Uber's operational performance.

Recommendation 1: Improve Driver Availability

Additional drivers should be allocated during periods of high demand to reduce failed bookings and improve ride completion rates.

Recommendation 2: Reduce Ride Cancellations

A detailed investigation into cancellation reasons should be conducted. Measures such as improved ride matching and better communication can help reduce cancellation rates.

Recommendation 3: Increase Customer Retention

Reward programs, discounts, and personalized offers can encourage customers to continue using the platform after their first ride.

Recommendation 4: Focus on High-Value Routes

Routes that consistently generate higher revenue should receive special attention to ensure reliable service and driver availability.

Recommendation 5: Enhance Customer Experience

Reducing wait times and improving booking reliability can increase customer satisfaction and strengthen brand loyalty.

Recommendation 6: Continuous Monitoring

Key performance indicators such as ride completion rate, cancellation rate, customer retention, and revenue should be monitored regularly using business intelligence dashboards.

Conclusion

This project explored Uber ride booking data to better understand ride performance, customer behavior, and revenue trends. Through the use of Python, SQL, and Power BI, several important insights were discovered regarding cancellations, customer retention, and operational efficiency.

The analysis showed that repeat customers play a major role in revenue generation, while ride cancellations and booking failures remain important challenges. By implementing the recommendations provided in this report, Uber can improve service quality, increase customer satisfaction, and enhance overall business performance.

This project demonstrates practical skills in data cleaning, exploratory data analysis, SQL querying, dashboard development, data visualization, and business decision-making.